

Stuart Lance Richards-Weir

Technologies and Languages

- Programming Languages: Java, JavaScript + TypeScript, Python, C, C++, SQL, Bash
- Backend & Systems: RESTful APIs, HTTP request lifecycle, service-to-service communication, data modelling
- Distributed Systems: AWS, Kafka, Spark, Docker, HDFS
- Tools: Git, Linux, Eclipse, Visual Studio
- Languages: English – Native Spanish – B1

Work Experience

Software Development Engineer Intern **Amazon** **July 2026 – December 2026 (Incoming)**
Payroll team Barcelona, Spain

- Selected for SDE Internship on Amazon's **Payroll Platform Engineering team** supporting payroll processing for millions of employees across multiple regulatory jurisdictions while utilising **AWS systems**.
- Contribute to a greenfield **internal platform** providing foundational payroll and payment-processing services **consumed by multiple engineering teams**.
- Design and implement **Java-based microservices** and **service APIs** with emphasis on contract stability, reliability and internal integration workflows.
- Participate in **system design** discussions around scalability, failure handling and service boundary definition for cross-team consumption.
- Extend and maintain a **TypeScript SDK** to standardise and simplify internal service adoption across teams.
- Help define **data models** and **service contracts** ensuring safe evolution of internal APIs.
- Contribute production-quality code following Amazon's design review and operational standards.

Software Engineer Intern **Dynatrace** **November 2025 – Current**
Data Visualisation and Dashboard teams Barcelona, Spain

- Worked on highly performant, interactable and customisable **telemetry-based data visualisation tools** using **React** and **TypeScript**, supporting up to **one million datapoints**.
- Designed and implemented a closest-point **search algorithm** for quick data selection, taking **5 FPS to 60+ (12x speedup)**
- Participated in **pair programming** and **code reviews**, incorporating feedback to improve correctness, readability and maintainability of **production code**.
- Collaborated in SCRUM-based Agile methodology at **enterprise level**, coordinating across teams and departments.

Education

- **B.Sc. Computer Engineering**, Universitat Pompeu Fabra, Spain. **Graduating 2026**
- **B.Sc. Mathematics and Physics**, University of Warwick, UK. (2 of 3 year course) **2020–2022**

Projects and Coursework

- **Undergraduate Thesis** - The Graph Isomorphism Problem: Algorithmic Optimisations for Scalable Graph Comparisons (in progress): Applications to distributed graph processing and large-scale network analysis.
- Implemented a **Python-based HDFS** client-server model for **distributed data storage and retrieval**, simulating **large-scale data transfer architecture** and **fault-tolerant data transfers** across nodes
- Built **MapReduce-based Spark jobs** to analyse tens of thousands of records; Utilised **Docker** containers and **AWS** separately for deploying jobs and collating results
- **Top 5 team**, from over 50, in **adversarial multi-agent Pacman AI competition** – Used heuristic search with a weighted A* algorithm to optimise **decision making under dynamic conditions**

About me

Computer Engineering student with a strong foundation in **Mathematics and Physics**. Eager to contribute to **large-scale, customer-obsessed** engineering challenges at the intersection of **distributed systems** and **system infrastructure**. Comfortable with fast-paced, ambiguous environments where creative problem-solving drives results. Currently taking **Spanish language classes** in order to better integrate into life in Spain.